AIML4OS

Artificial Intelligence and Machine Learning for Official Statistics

Newsletter 4

Welcome to our Newsletter #4 for ESSnet AIML4OS! Here you will find project updates highlighting progress, achievements, events and all news.

PROJECT OVERVIEW

The main objectives of AIML4OS are to explore the use of Artificial Intelligence/Machine Learning (AI/ML) for the production of official statistics and to implement innovative solutions for statistical products and processes. This four-year project started in April 2024, with activities structured in the following work packages

OVERARCHING WORK PACKAGES

WP1
Project management and coordination

WP2
Communication and community engagement

WP3
ESS AI/ML lab: Technical infrastructure and organisational setup

WP4
AI/ML state-of-play and ecosystem monitoring

WP5
Standards, methodological and implementation frameworks

WP6
Knowledge repository and training material

USE CASES

WP7
AI/ML on earth observation data, satellite imagery

WP8
Statistically valid and efficient editing and imputation in official statistics by AI/ML – with a special focus on editing

WP9
Imputation focus - Statistically valid and efficient editing and imputation in official statistics by AI/ML – with a special focus on imputation

WP10
From text to code - Experiences and potential of the use of AI/ML for classifying and coding

WP11
Applying ML for estimating firm-level supply chain networks

WP12
Large language models

WP13
Generation of synthetic data in official statistics: techniques and applications

In this issue the Hackathon for Large Language Models, From Text to Code and Imputation task progress & the Supply Chain Network achievements

READ MORE...

LLM AT WORK: EVALUATING AND PROTOTYPING AI FOR STATISTICAL IMPACT

Large language models (WP12)

A two-day hackathon was held in **Lisbon, June 2025**, with participation from project members representing **Portugal, the Netherlands, Sweden, Ireland, and Norway**, and with **technical support from France**, who delivers the shared data lab infrastructure (SSPCloud) used in the AIML4OS project.



Team Members
WP12 - AIML4OS

The event began with a productive discussion focused on identifying high-value business areas, particularly in the domains of data collection and data dissemination. From there, the group worked on establishing a set of principles to evaluate the potential of each idea put forward.
These principles were developed with the rapid advancement of large language models (LLMs) in mind, and include:

- **Data accessibility** – considering both sensitivity and availability
- **Reusability** – value beyond a single use case, ideally benefiting multiple statistical offices
- **Feasibility** – practical implementation within existing constraints
- **Efficiency gain** – potential to streamline processes or reduce workload
- **Life span** – expected relevance before underlying conditions change
- **On-premises compatibility** – suitability for deployment within NSIs' infrastructure
- **Low-hanging fruit for NSIs** – ease of implementation and quick wins
- **Evaluation** – ability to verify and assess the outcomes

Based on these principles, the group selected the most promising ideas for prototyping:

Prototype A. Web Corner Prototype- Explore how modern web scraping techniques, combined with Large Language Models (LLMs), can pinpoint specific content on webpages. **Goal:** To determine if a webpage contains information related to a specific variable or concept. For example: "Does this webpage contain a job vacancy?", "Is this website an accommodation site?"

Prototype B. Dissemination Summary Prototype - Solve a common need in statistical dissemination workflows: summarizing reports in both local languages (like Portuguese) and English, along with relevant keywords or tags. **Goal:** To evaluate if LLMs can efficiently support and streamline this summarization process, ensuring reusability, transferability, and compatibility with statistical organizations' requirements.

Prototype C. From PDF to Figures - Investigate how AI tools can extract and process structured quantitative content (tables, figures, numeric summaries) from PDF-formatted annual reports and similar documents. **Goal:** To enable the downstream use of this information for analysis, data validation, or integration into other statistical processes.

While these initial versions were simple, they hold significant potential for future expansion and development. If you would like to check out the ongoing work with the prototypes – check the workpackage repo!







Team Members WP12 AIML4OS

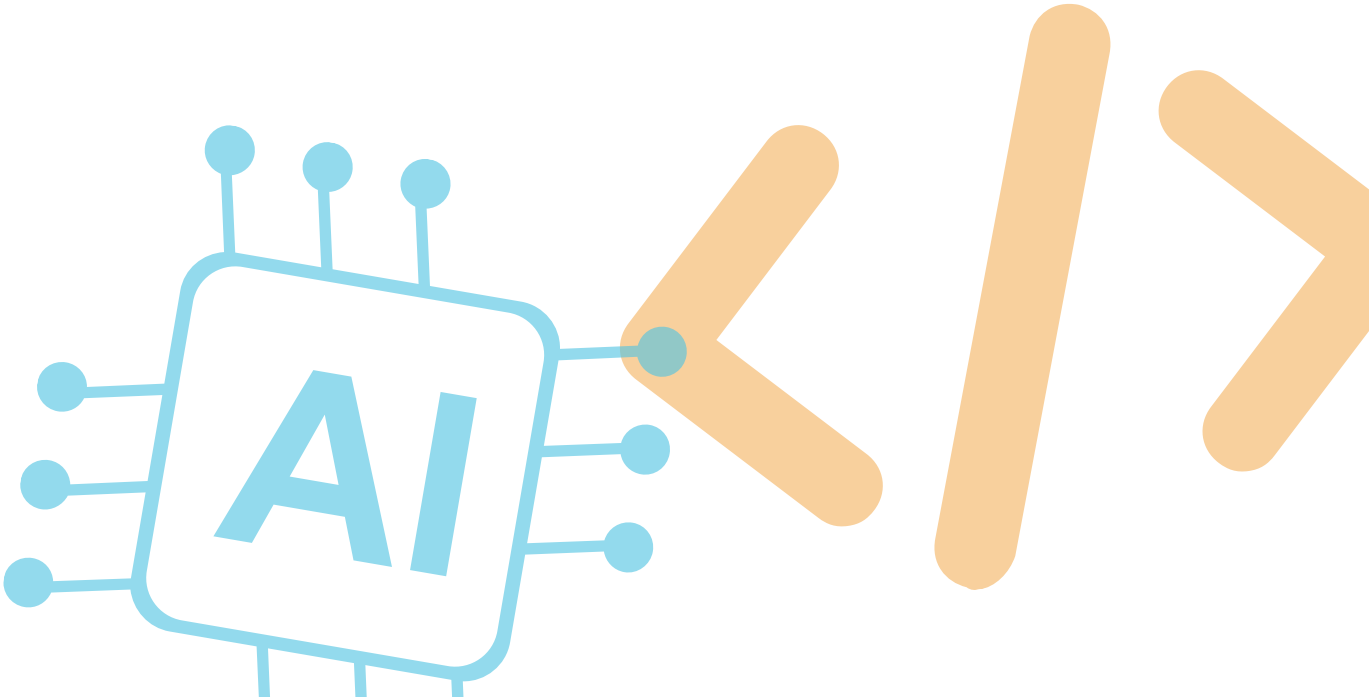
BEYOND THE LABEL: BUILDING BETTER TEXT CLASSIFIERS FOR OFFICIAL DATA

From text to code (WP10)

The team is nearing the completion of its first task, which aimed to provide an **overview of text classification** using machine learning within the context of official statistics, along with a literature review on the topic. They will host the final two presentations—from the Netherlands and Germany—bringing the total to **11 participating countries** that have presented their projects on this topic. These presentations have provided valuable insights into how different statistical offices are approaching text classification. Based on this input, Germany is currently drafting a summary report.
As part of this task, the participating countries also conducted literature reviews on five key topics:

- Exploring the **potential of using synthetic data** for developing NLP classifiers
- Using LLMs and **Retrieval-Augmented Generation** (RAG) for text classification
- Implementing an **MLOps approach** for production-grade machine learning
- **Hierarchical methods** for text classification
- **Addressing misclassified labeled** data in text classification tasks

Each group has produced a concise summary paper intended to support future NSI Data Scientists working on these topics. These **papers will soon be available** on GitHub!



MILESTONE ACHIEVED: PORTUGUESE FIRM-LEVEL SUPPLY CHAIN NETWORK COMPLETED

Applying ML for estimating firm-level supply chain networks (WP11)

The team finished its **first output**! Based on very rich **Portuguese administrative data**, a firm-level supply chain network was created for 2022. This will be used in the project for **deriving training and test sets** for training models for **reconstructing firm-level networks** for other countries. **A report** on this part of the project was **delivered to Eurostat**. The dataset will also become **available for research** on-site at Statistics Portugal.

The AIML4OS project and the team **work were presented at the international NetSci 2025 conference** in June, organized by the Network Science Society. The reactions were quite positive and sparked interest in the academic world.

FIRST AIML4OS TRAINING MATERIAL TEMPLATE NOW AVAILABLE

Knowledge repository and training material (WP6)

The team has been preparing the **first training material** in a form of slides. It has two purposes:

- To **open the floor** of our training repository.
- To **serve as baseline** scenario **for other training material** since the work establishing many technical parameters to publish it on GitHub pages is already done.

The **training material is available** at the following link



STRENGTHENING COLLABORATION: STANDARDIZATION AND KNOWLEDGE SHARING UNDERWAY

Imputation focus (WP9)

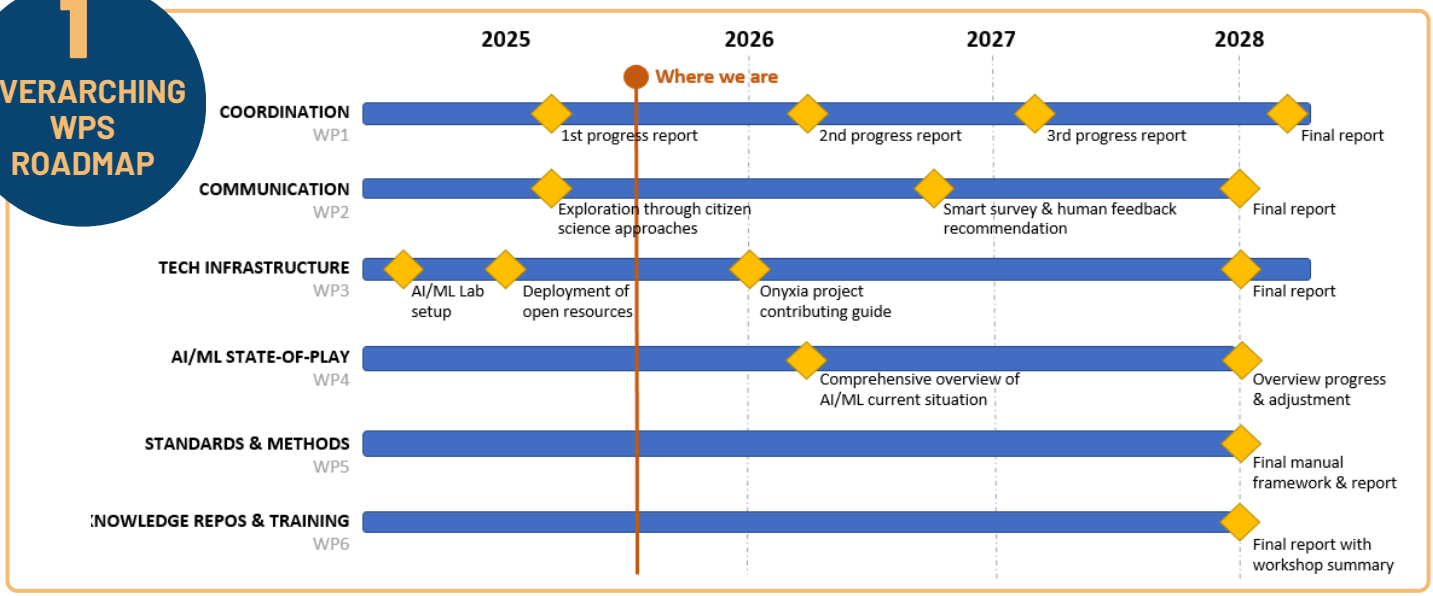
Good progress on the **work across the three subgroups' projects**. Started to standardising information to share **using a common template**. This will serve as the foundation for the content of the deliverables. Furthermore, the **creation of GitHub repositories** for each project has begun, and they will start sharing code during meetings so that everyone can learn from the developments and implementations carried out by other participants. Another important development has been the **collaboration with Standards, methodological and implementation frameworks team (WP5)**, focusing on methodological aspects. We have shared potential areas for collaboration and established direct contact, who presented the Total Machine Learning Error Model. This model will serve as a foundation for our deliverable on quality.

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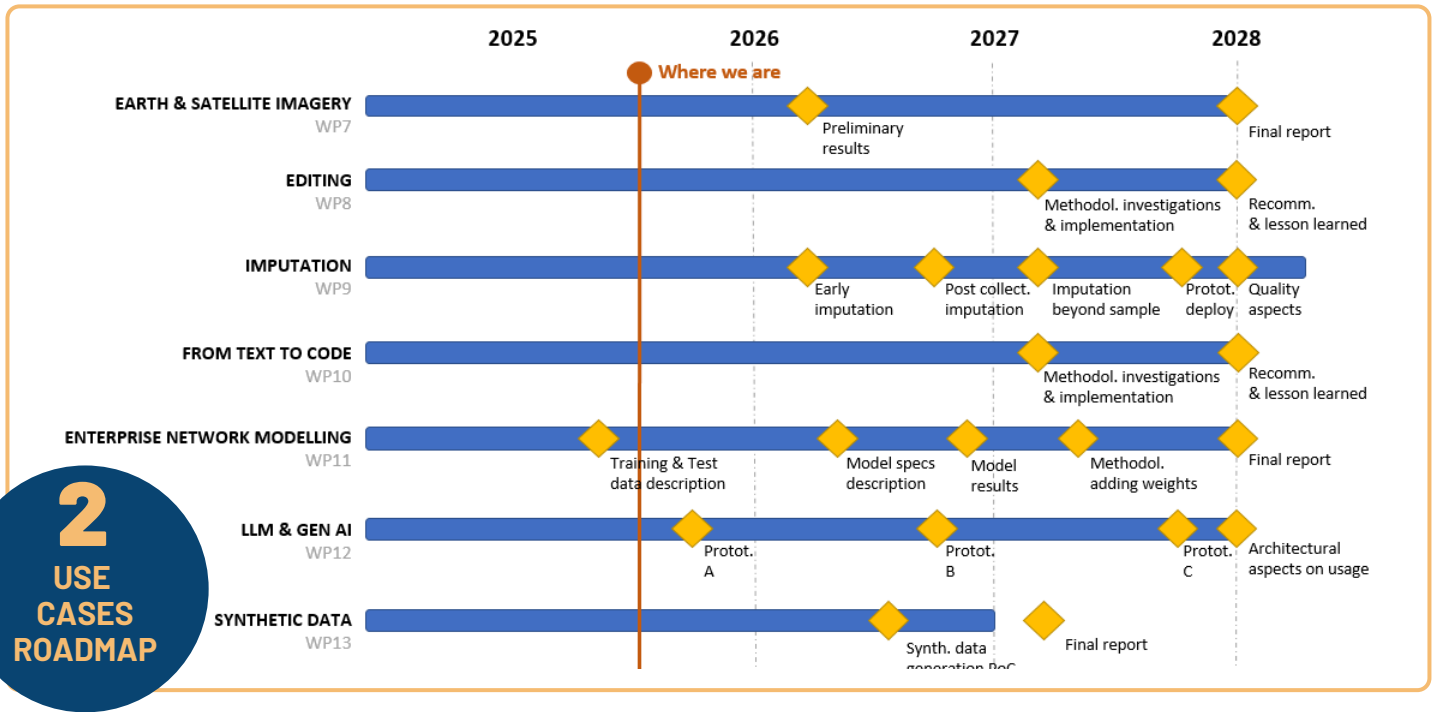
For each WP involved, the project is divided into several phases. Below are descriptions of what will be achieved and when.

PROJECT OVERVIEW

1 OVERARCHING WPS ROADMAP



2 USE CASES ROADMAP



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